ASSIGNMENT-14.1

HT NO:2303A51585

Batch No:2303A51585

Course:AI Assisted Coding

Task 1 : Portfolio Website Design

You are building a personal portfolio website to showcase your

work.

Requirements:

• Create sections for About Me, Projects, and Contact.

• Use AI to:

o Suggest color palettes and typography.

o Create a responsive layout with Grid/Flexbox.

o Add smooth scrolling navigation.

Prompt:

Generate a complete single-file HTML code for a personal portfolio website that I can run in VS Code.

Requirements:

- Sections: About Me, Projects, Contact

- Use CSS Flexbox or Grid for layout

- Suggest a modern color palette and typography

- Add a navigation bar with smooth scrolling between sections

- Keep HTML, CSS, and JavaScript in the same file

- The code should be clean and easy for beginners

Code:

```html
<!--Generate a complete single-file HTML code for a personal portfolio website that I can run in VS Code

Requirements:
- Sections: About Me, Projects, Contact
- Use CSS Flexbox or Grid for layout
- Suggest a modern color palette and typography
- Add a navigation bar with smooth scrolling between sections
- Keep HTML, CSS, and JavaScript in the same file
- The code should be clean and easy for beginners-->
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>My Portfolio</title>
        <style>
            /* Reset some default styles */
            * {
                margin: 0;
                padding: 0;
                box-sizing: border-box;
            }

            body {
                font-family: 'Arial', sans-serif;
                background-color: #f4f4f9;
                color: #333;
            }
```

```css
/* Navigation Bar */
nav {
    background-color: #4CAF50;
    padding: 1rem;
    position: fixed;
    width: 100%;
    top: 0;
    left: 0;
    z-index: 1000;
}

nav ul {
    display: flex;
    justify-content: center;
    list-style-type: none;
}

nav ul li {
    margin: 0 15px;
}

nav ul li a {
    color: white;
    text-decoration: none;
    font-size: 1.2rem;
```

```
        text-decoration: none;
        font-size: 1.2rem;
    }

    nav ul li a:hover {
        text-decoration: underline;
    }

    /* Sections */
    section {
        padding: 100px 20px; /* Add padding to account for fixed nav */
        min-height: 100vh; /* Full viewport height */
    }

    #about {
        background-color: #e3f2fd;
    }

    #projects {
        background-color: #c8e6c9;
    }

    #contact {
        background-color: #ffccbc;
    }
```

```
    }

    /* Layout using Flexbox */
    .container {
        display: flex;
        flex-direction: column;
        align-items: center;
        max-width: 1200px;
        margin: auto;
    }

    h1, h2 {
        margin-bottom: 20px;
    }

    p {
        max-width: 600px;
        text-align: center;
        margin-bottom: 20px;
    }
</style>
</head>
<body>
    <nav>
        <ul>
```

```html
<html lang="en">
    <body>
        <nav>
            <ul>
                <li><a href="#about">About Me</a></li>
                <li><a href="#projects">Projects</a></li>
                <li><a href="#contact">Contact</a></li>
            </ul>
        </nav>

        <section id="about">
            <div class="container">
                <h1>About Me</h1>
                <p>Hello! I'm a passionate developer with experience in web development, design, and programming.
                    I love creating beautiful and functional websites that provide great user experiences.</p>
            </div>
        </section>

        <section id="projects">
            <div class="container">
                <h2>Projects</h2>
                <p>Here are some of my recent projects:</p>
                <ul>
                    <li>Project 1: A responsive portfolio website built with HTML, CSS, and JavaScript.</li>
                    <li>Project 2: A task management app using React and Node.js.</li>
                    <li>Project 3: A personal blog created with Gatsby and GraphQL.</li>
                </ul>
            </div>
        </section>
```

```html
        </section>

        <section id="contact">
            <div class="container">
                <h2>Contact</h2>
                <p>If you'd like to get in touch, feel free to reach out via email or connect with me on LinkedIn!</p>
                <p>Email: <a href="mailto:></p>
                <p>LinkedIn: <a href="https://www.linkedin.com/in/yourprofile" target="_blank">Your LinkedIn Profile</a></p>
            </div>
        </section>
        <script>
            // Smooth scrolling for navigation links
            document.querySelectorAll('nav ul li a').forEach(anchor => {
                anchor.addEventListener('click', function (e) {
                    e.preventDefault();
                    document.querySelector(this.getAttribute('href')).scrollIntoView({
                        behavior: 'smooth'
                    });
                });
            });
        </script>
    </body>
</html>
```

Output:

About Me Projects Contact

About Me

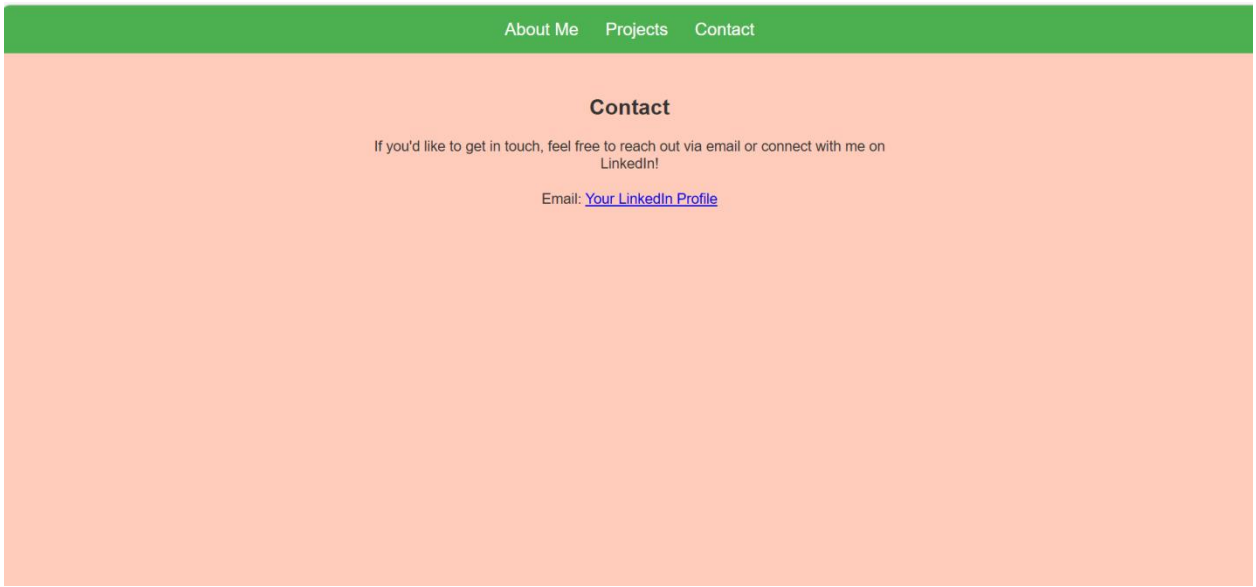
Hello! I'm a passionate developer with experience in web development, design, and programming. I love creating beautiful and functional websites that provide great user experiences.

About Me Projects Contact

Projects

Here are some of my recent projects:

- Project 1: A responsive portfolio website built with HTML, CSS, and JavaScript.
- Project 2: A task management app using React and Node.js.
- Project 3: A personal blog created with Gatsby and GraphQL.



Justification:

1. The portfolio website helps showcase personal skills, projects, and contact details in a structured way.

2. Separate sections like About, Projects, and Contact improve organization and readability.

3. Using Flexbox or Grid creates a responsive layout that adapts to different screen sizes.

4. A consistent color palette and typography improve visual appeal and professionalism.

5. Smooth scrolling navigation enhances user experience.

6. Overall, the design makes the portfolio easy to navigate and visually attractive.

Task 2 :Online Store Product Page

Design a product display page for an online store.

Requirements:

• Display product image, title, price, and "Add to Cart" button.

• Use AI to:

o Style with BEM methodology.

o Make layout responsive.

o Add hover effects and "Add to Cart" alert.

Prompt:

Generate a single HTML file for a responsive online store product page that can run in VS Code.

Requirements:

- Display product image, title, price, and an "Add to Cart" button

- Use BEM naming methodology in CSS classes

- Add hover effects on the product card and button

- When the "Add to Cart" button is clicked, show a JavaScript alert

- Include HTML, CSS, and JavaScript in the same file

Code:

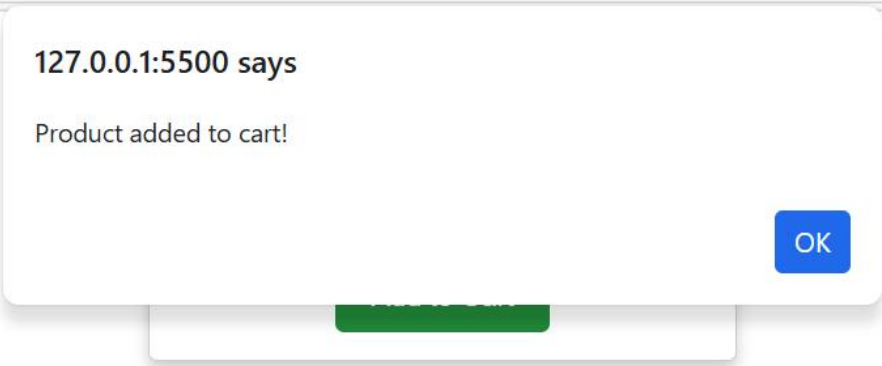
```html
<!--Generate a single HTML file for a responsive online store product page that can run in VS Code.

Requirements:
- Display product image, title, price, and an "Add to Cart" button
- Use BEM naming methodology in CSS classes
- Add hover effects on the product card and button
- When the "Add to Cart" button is clicked, show a JavaScript alert
- Include HTML, CSS, and JavaScript in the same file-->
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Responsive Online Store Product Page</title>
        <style>
            /* CSS Styles */
            .product-card {
                border: 1px solid #ccc;
                border-radius: 5px;
                padding: 16px;
                max-width: 300px;
                margin: auto;
                text-align: center;
                transition: box-shadow 0.3s ease;
            }
            .product-card:hover {
                box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
            }
            .product-card__image {
```

```
        .product-card__image {
            width: 100%;
            height: auto;
            border-radius: 5px;
        }
        .product-card__title {
            font-size: 1.5em;
            margin: 10px 0;
        }
        .product-card__price {
            color: #888;
            font-size: 1.2em;
            margin-bottom: 20px;
        }
        .product-card__button {
            background-color: #28a745;
            color: white;
            border: none;
            padding: 10px 20px;
            font-size: 1em;
            cursor: pointer;
            border-radius: 5px;
            transition: background-color 0.3s ease;
        }
        .product-card__button:hover {
            background-color: #218838;
```

```
        .product-card__button:hover {
        }
    </style>
    <script>
        // JavaScript Function
        function addToCart() {
            alert("Product added to cart!");
        }
    </script>
    <script>
        // Ensure the DOM is fully loaded before attaching event listeners
        document.addEventListener("DOMContentLoaded", function() {
            const addToCartButton = document.querySelector(".product-card__button");
            addToCartButton.addEventListener("click", addToCart);
        });
    </script>

</head>
<body>
    <!-- HTML Structure -->
    <div class="product-card">
        <img src="https://via.placeholder.com/300" alt="Product Image" class="product-card__image">
        <h2 class="product-card__title">Sample Product</h2>
        <p class="product-card__price">$29.99</p>
        <button class="product-card__button">Add to Cart</button>
    </div>
```

```html
    </head>
    <body>
        <!-- HTML Structure -->
        <div class="product-card">
            <img src="https://via.placeholder.com/300" alt="Product Image" class="product-card__image">
            <h2 class="product-card__title">Sample Product</h2>
            <p class="product-card__price">$29.99</p>
            <button class="product-card__button">Add to Cart</button>
        </div>
    </body>
</html>
```

Output:



Justification:

1. The product page displays essential product information such as image, title, and price.

2. The "Add to Cart" button allows basic interaction for users.

3. Using BEM methodology improves CSS structure and maintainability.

4. Responsive design ensures the page works on both mobile and desktop devices.

5. Hover effects provide visual feedback when users interact with elements.

6. The alert message confirms that the product has been added to the cart.

Task 3 :Event Registration Form

Build an event registration form for a conference.

Requirements:

• Collect name, email, phone number, and session selection.

• Use AI to:

o Add form validation with JavaScript.

o Make the form accessible with labels and ARIA.

o Style with a professional look

Prompt:

Generate a single HTML file for an event registration form that I can run in VS Code.

Requirements:

- Fields: Name, Email, Phone Number, Session Selection

- Add JavaScript form validation

- Use labels for accessibility and simple ARIA attributes

- Style the form with a professional design

- Include HTML, CSS, and JavaScript in one file

Code:

```html
<!--Generate a single HTML file for an event registration form that I can run in VS Code.

Requirements:
- Fields: Name, Email, Phone Number, Session Selection
- Add JavaScript form validation
- Use labels for accessibility and simple ARIA attributes
- Style the form with a professional design
- Include HTML, CSS, and JavaScript in one file-->
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Event Registration Form</title>
        <style>
            body {
                font-family: Arial, sans-serif;
                background-color: #f4f4f4;
                display: flex;
                justify-content: center;
                align-items: center;
                height: 100vh;
                margin: 0;
            }
            .registration-form {
                background-color: #fff;
                padding: 20px;
                border-radius: 5px;
                box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
```

```
                width: 300px;
            }
            .registration-form h2 {
                margin-bottom: 20px;
            }
            .form-group {
                margin-bottom: 15px;
            }
            .form-group label {
                display: block;
                margin-bottom: 5px;
            }
            .form-group input, .form-group select {
                width: 100%;
                padding: 8px;
                box-sizing: border-box;
            }
            .submit-btn {
                background-color: #007BFF;
                color: #fff;
                border: none;
                padding: 10px 15px;
                cursor: pointer;
                border-radius: 5px;
            }
```

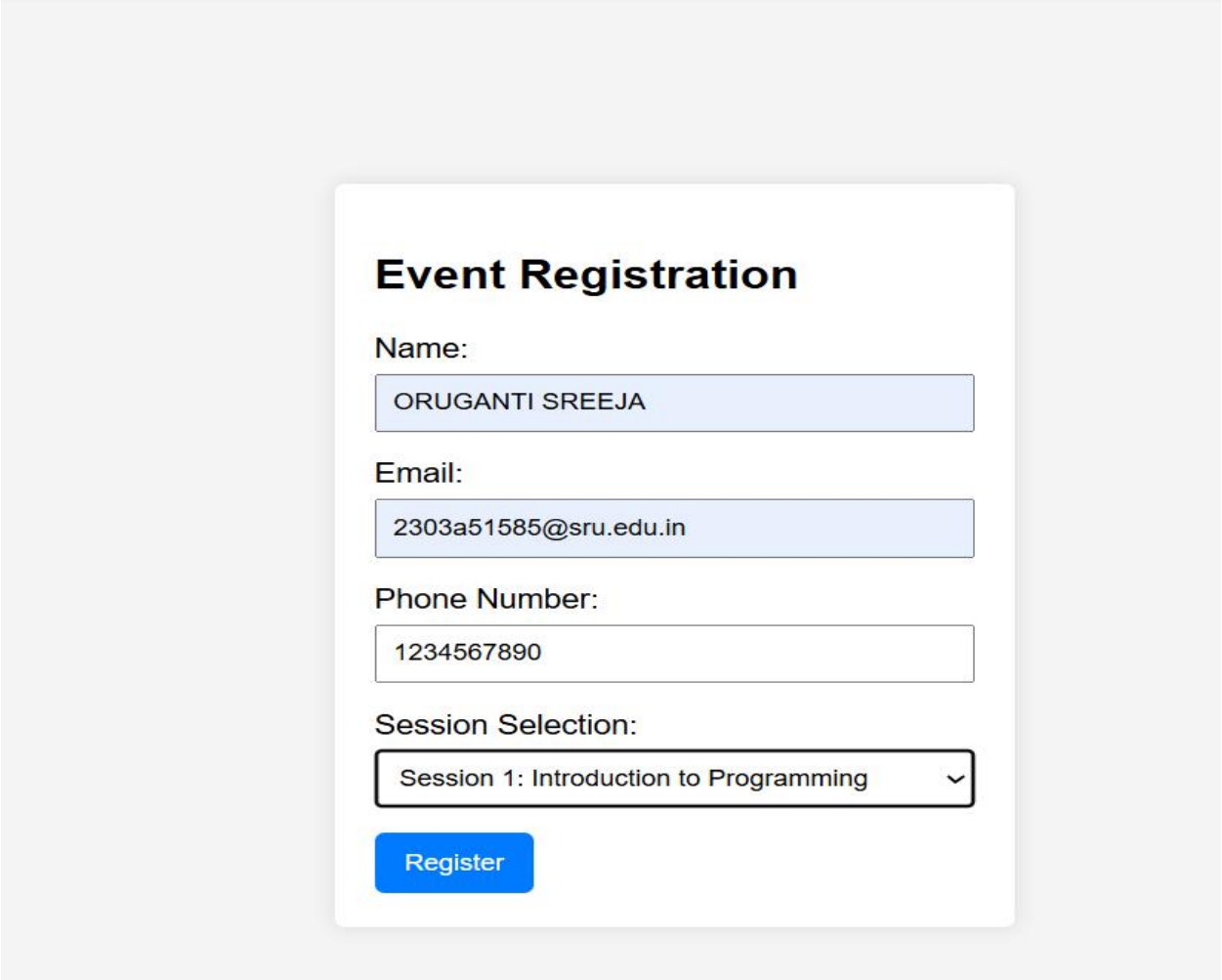
```
.submit-btn:hover {
    background-color: #0056b3;
}
</style>
<script>
    function validateForm() {
        const name = document.getElementById('name').value.trim();
        const email = document.getElementById('email').value.trim();
        const phone = document.getElementById('phone').value.trim();
        const session = document.getElementById('session').value;

        if (name === '') {
            alert('Please enter your name.');
            return false;
        }
        if (email === '') {
            alert('Please enter your email.');
            return false;
        }
        if (!validateEmail(email)) {
            alert('Please enter a valid email address.');
            return false;
        }
        if (phone === '') {
            alert('Please enter your phone number.');
            return false;
```

```
        }
        if (!validatePhone(phone)) {
            alert('Please enter a valid phone number.');
            return false;
        }
        if (session === '') {
            alert('Please select a session.');
            return false;
        }
        return true;
    }

    function validateEmail(email) {
        const re = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
        return re.test(email);
    }

    function validatePhone(phone) {
        const re = /^\d{10}$/;
        return re.test(phone);
    }
</script>
</head>
<body>
    <form class="registration-form" onsubmit="return validateForm()">
```

```html
            <h2>Event Registration</h2>
            <div class="form-group">
                <label for="name">Name:</label>
                <input type="text" id="name" name="name" aria-required="true">
            </div>
            <div class="form-group
">
                <label for="email">Email:</label>
                <input type="email" id="email" name="email" aria-required="true">
            </div>
            <div class="form-group">
                <label for="phone">Phone Number:</label>
                <input type="text" id="phone" name="phone" aria-required="true">
            </div>
            <div class="form-group">
                <label for="session">Session Selection:</label>
                <select id="session" name="session" aria-required="true">
                    <option value="">--Select a session--</option>
                    <option value="session1">Session 1: Introduction to Programming</option>
                    <option value="session2">Session 2: Advanced JavaScript</option>
                    <option value="session3">Session 3: Web Development Best Practices</option>
                </select>
            </div>
            <button type="submit" class="submit-btn">Register</button>
        </form>
    </body>
```


```html
                <option value="session1">Session 1: Introduction to Programming</option>
                <option value="session2">Session 2: Advanced JavaScript</option>
                <option value="session3">Session 3: Web Development Best Practices</option>
            </select>
        </div>
        <button type="submit" class="submit-btn">Register</button>
    </form>
</body>

</html>
```


Output:



Justification:

1. The registration form collects important participant details for an event.

2. Fields like name, email, phone number, and session help organize attendee information.

3. JavaScript validation ensures that users enter correct and required data.

4. Using labels and ARIA attributes improves accessibility for all users.

5. A clean and professional design makes the form easy to use.

6. This improves both usability and data accuracy.

Task 4 : Resume Website

Build a one-page online resume.

Requirements:

• Sections: Profile Summary, Skills, Education, Experience.

• Use AI to:

o Propose professional typography styles.

o Align sections using CSS Grid.

o Add smooth scroll navigation between sections.

Prompt:

Generate a single HTML file for a one-page resume website that can run in VS Code.

Requirements:

- Sections: Profile Summary, Skills, Education, Experience

- Use professional typography

- Align sections using CSS Grid

- Add a navigation menu with smooth scrolling

- Include HTML, CSS, and JavaScript in the same file

Code:

```html
<!--Generate a single HTML file for a one-page resume website that can run in VS Code.

Requirements:
- Sections: Profile Summary, Skills, Education, Experience
- Use professional typography
- Align sections using CSS Grid
- Add a navigation menu with smooth scrolling
- Include HTML, CSS, and JavaScript in the same file-->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Resume</title>
    <style>
        body {
            font-family: 'Arial', sans-serif;
            margin: 0;
            padding: 0;
            background-color: #f4f4f4;
        }
        header {
            background-color: #333;
            color: #fff;
            padding: 10px 0;
            text-align: center;
        }
        nav {
            display: flex;
```

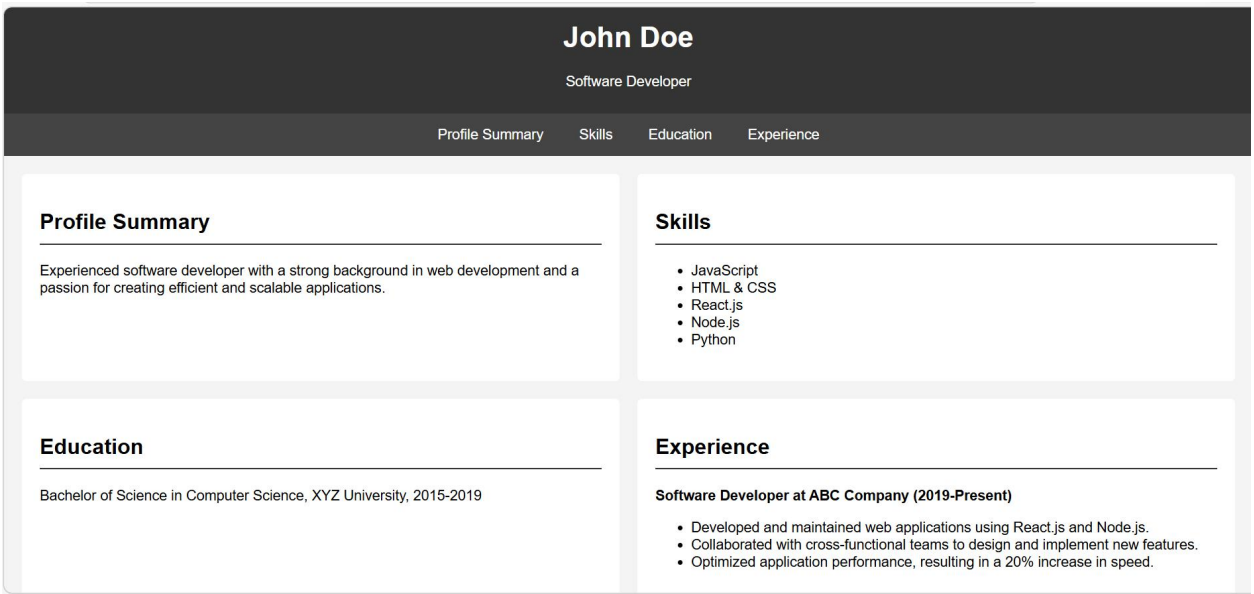
```html
<html lang="en">
<head>
    <style>
        nav {
            justify-content: center;
            background-color: #444;
        }
        nav a {
            color: #fff;
            padding: 14px 20px;
            text-decoration: none;
        }
        nav a:hover {
            background-color: #555;
        }
        .container {
            display: grid;
            grid-template-columns: 1fr 1fr;
            gap: 20px;
            padding: 20px;
        }
        section {
            background-color: #fff;
            padding: 20px;
            border-radius: 5px;
        }
        h2 {
            border-bottom: 2px solid #333;
            padding-bottom: 10px;
```

```
        }
    </style>
</head>
<body>
    <header>
        <h1>John Doe</h1>
        <p>Software Developer</p>
    </header>
    <nav>
        <a href="#profile">Profile Summary</a>
        <a href="#skills">Skills</a>
        <a href="#education">Education</a>
        <a href="#experience">Experience</a>
    </nav>
    <div class="container">
        <section id="profile">
            <h2>Profile Summary</h2>
            <p>Experienced software developer with a strong background in web development and a
                passion for creating efficient and scalable applications.</p>
        </section>
        <section id="skills">
            <h2>Skills</h2>
            <ul>
                <li>JavaScript</li>
                <li>HTML & CSS</li>
                <li>React.js</li>
```

```
                <li>Node.js</li>
                <li>Python</li>
            </ul>
        </section>
        <section id="education">
            <h2>Education</h2>
            <p>Bachelor of Science in Computer Science, XYZ University, 2015-2019</p>
        </section>
        <section id="experience">
            <h2>Experience</h2>
            <p><strong>Software Developer at ABC Company (2019-Present)</strong></p>
            <ul>
                <li>Developed and maintained web applications using React.js and Node.js.</li>
                <li>Collaborated with cross-functional teams to design and implement new features.</li>
                <li>Optimized application performance, resulting in a 20% increase in speed.</li>
            </ul>
        </section>
    </div>

    <script>
        // Smooth scrolling for navigation links
        document.querySelectorAll('nav a').forEach(anchor => {
            anchor.addEventListener('click', function(e) {
```

```html
    <script>
        // Smooth scrolling for navigation links
        document.querySelectorAll('nav a').forEach(anchor => {
            anchor.addEventListener('click', function(e) {
                e.preventDefault();
                document.querySelector(this.getAttribute('href')).scrollIntoView({
                    behavior: 'smooth'
                });
            });
        });
    </script>
</body>
</html>
```

Output:



Justification:

1. The one-page resume website presents professional information in a structured format.
2. Sections like Profile, Skills, Education, and Experience help organize content clearly.
3. Professional typography improves readability and presentation.
4. CSS Grid helps align sections neatly across the page.
5. Smooth scrolling navigation allows users to quickly move between sections.
6. This format creates a modern and accessible online resume.

Task 5 : Restaurant Menu Page

Design a digital menu page for a restaurant.

Requirements:

• Display menu items with name, description, and price.

• Use AI to:

o Suggest a food-friendly color palette.

o Design a responsive card-based layout.

o Add hover effects on menu items.

Prompt:

Generate a single HTML file for a restaurant digital menu page that I can run in VS Code.

Requirements:

- Show menu items with name, description, and price

- Use a food-friendly color palette

- Design a responsive card-based layout

- Add hover effects to menu cards

- Include HTML and CSS in the same file

Code:

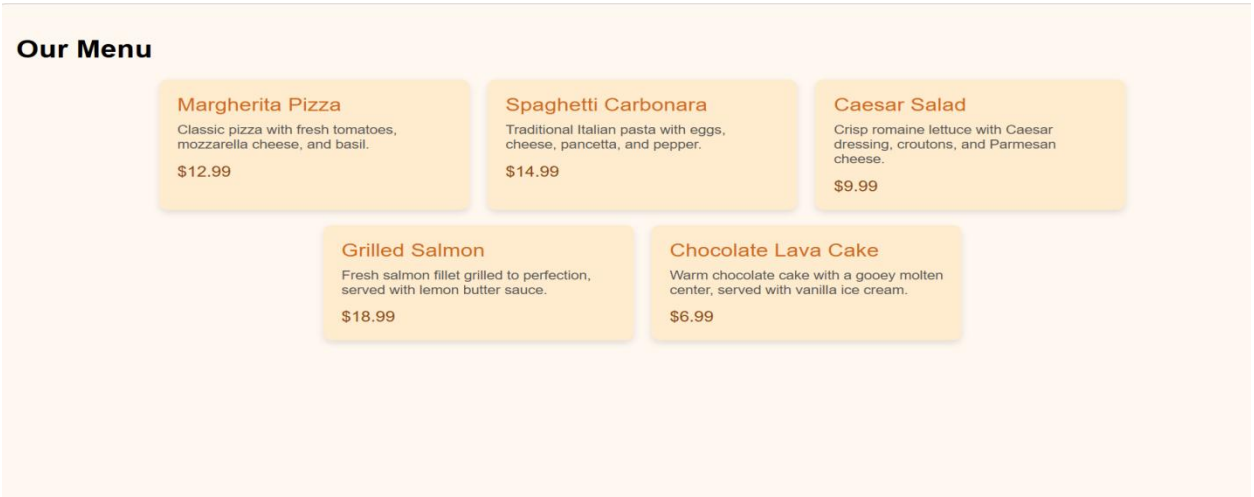
```html
<!--Generate a single HTML file for a restaurant digital menu page that I can run in VS Code.
Requirements:
- Show menu items with name, description, and price
- Use a food-friendly color palette
- Design a responsive card-based layout
- Add hover effects to menu cards
- Include HTML and CSS in the same file-->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Restaurant Digital Menu</title>
    <style>
        body {
            font-family: 'Arial', sans-serif;
            background-color: #fff8f0;
            margin: 0;
            padding: 20px;
        }
        .menu-container {
            display: flex;
            flex-wrap: wrap;
            gap: 20px;
            justify-content: center;
        }
        .menu-card {
            background-color: #ffebcd;
            border-radius: 10px;
```

```
            box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
            width: 300px;
            padding: 20px;
            transition: transform 0.3s ease, box-shadow 0.3s ease;
        }
        .menu-card:hover {
            transform: translateY(-10px);
            box-shadow: 0 8px 16px rgba(0, 0, 0, 0.2);
        }
        .menu-item-name {
            font-size: 1.5em;
            color: #d2691e;
            margin-bottom: 10px;
        }
        .menu-item-description {
            font-size: 1em;
            color: #555;
            margin-bottom: 15px;
        }
        .menu-item-price {
            font-size: 1.2em;
            color: #8b4513;
        }
    </style></head>
<body>
```

```html
<body>
    <h1>Our Menu</h1>
    <div class="menu-container">
        <div class="menu-card">
            <div class="menu-item-name">Margherita Pizza</div>
            <div class="menu-item-description">Classic pizza with fresh tomatoes, mozzarella cheese, and basil.</div>
            <div class="menu-item-price">$12.99</div>
        </div>
        <div class="menu-card">
            <div class="menu-item-name">Spaghetti Carbonara</div>
            <div class="menu-item-description">Traditional Italian pasta with eggs, cheese, pancetta, and pepper.</div>
            <div class="menu-item-price">$14.99</div>
        </div>
        <div class="menu-card">
            <div class="menu-item-name">Caesar Salad</div>
            <div class="menu-item-description">Crisp romaine lettuce with Caesar dressing, croutons, and Parmesan cheese.</div>
            <div class="menu-item-price">$9.99</div>
        </div>
        <div class="menu-card">
            <div class="menu-item-name">Grilled Salmon</div>
            <div class="menu-item-description">Fresh salmon fillet grilled to perfection, served with lemon butter sauce.</div>
            <div class="menu-item-price">$18.99</div>
        </div>
        <div class="menu-card">
            <div class="menu-item-name">Chocolate Lava Cake</div>
            <div class="menu-item-description">Warm chocolate cake with a gooey molten center, served with vanilla ice cream.</di
            <div class="menu-item-price">$6.99</div>
        </div>
```

```html
        </div>
        <div class="menu-card">
            <div class="menu-item-name">Grilled Salmon</div>
            <div class="menu-item-description">Fresh salmon fillet grilled to perfection, served with lemon butter sauce.</div>
            <div class="menu-item-price">$18.99</div>
        </div>
        <div class="menu-card">
            <div class="menu-item-name">Chocolate Lava Cake</div>
            <div class="menu-item-description">Warm chocolate cake with a gooey molten center, served with vanilla ice cream.</di
            <div class="menu-item-price">$6.99</div>
        </div>
    </div>
</body>
</html>
```

Output:



Justification:

1. The digital menu page allows restaurants to present menu items online.
2. Each item includes name, description, and price for clear information.
3. A food-friendly color palette enhances the visual appeal of the page.
4. Card-based layout organizes menu items neatly and improves readability.
5. Hover effects provide interactive feedback for users.
6. Responsive design ensures the menu works well on phones, tablets, and desktops